

Assignment 1: Large Integer Arithmetic Expression

Objective: In this lab, you will implement a program that evaluates large integer arithmetic expressions with multiple operands and parentheses. You will work with large integers (up to 100 digits) and ensure the proper handling of basic arithmetic operations, respecting the precedence of operations, including the use of parentheses.

Problem Statement: Given a mathematical infix expression containing large integers (up to 100 digits) and multiple arithmetic operators (+, -, *, /), your task is to evaluate the expression and return the result. The expression may also include parentheses to group operations. Your program should handle expressions with multiple operands, correctly respecting the order of operations (PMDAS: Parentheses, Multiplication/Division, Addition/Subtraction).

Requirements:

1. Input:

- A string representing the arithmetic expression. The expression may contain:
 1. Large integers (up to 100 digits).
 2. The four basic arithmetic operators (+, -, *, /).
 3. Parentheses to denote precedence in operations.
- **Example Input:** (12345678901234567890 + (98765432109876543210 - 12345678901234567890)) * 100

2. Output:

- The result of evaluating the arithmetic expression, taking into account the proper order of operations, including the handling of parentheses.
- **Example Output:** 1111111111111111111100000000000000000000.

3. Tasks:

- Replace StudentID with your student ID.
- Take the input from the file name “tests.txt”. Output the results to the file named “output_StudentID.txt”.
- Each line in the file “tests.txt” is an arithmetic expression.
- Your program must be a C++ program. The cpp file containing the main function must be named with your student ID, i.e., StudentID.cpp, which takes **two command-line arguments** x and y where x=tests.txt and y=output_StudentID.txt, then prints the results of each expression in “tests.txt”:

For example, consider the case when `tests.txt` contains two expressions as follows:

$$\frac{(12345678901234567890 + (98765432109876543210 - 12345678901234567890)) * 100}{1 + 2 * 4}$$

When using the following command line in terminal or cmd (in Windows):

```
g++ StudentID.cpp -o StudentID
```

StudentID tests.txt output StudentID.txt (or ./StudentID tests.txt

output StudentID.txt)

the following results must be shown in the terminal or cmd and written in the file

4. Any violation of this guideline or the tasks will result in zero.